

HK IMO Training 2020-2021
Problem Set 13
For 30 January 2020



Problem 49. Find all triples (a, b, c) of nonnegative integers that satisfy

$$a! + 5^b = 7^c.$$

Problem 50. Find all integers $n > 1$ such that for each $r = 0, 1, 2, \dots, n-1$, there exists a multiple of n whose sum of digits leaves a remainder r when divided by n .

Problem 51. Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients. Let $a_n = n! + n$. Show that if $\frac{P(a_n)}{Q(a_n)}$ is an integer for every n , then $\frac{P(n)}{Q(n)}$ is an integer for every integer n such that $Q(n) \neq 0$.

Problem 52. Prove that there are infinitely many positive integers n such that n divides $2017^{2017^n-1} - 1$ but n does not divide $2017^n - 1$.